



USER MANUAL

MOBILE GEOGRAPHICAL INFORMATION SYSTEM

Version 1.0.0

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INTRODUCTION

General Information about the System

Mobile Geographical Information System is an application designed to bring the Web GIS interface to the mobile device and support data collection of ZETDC infrastructure. It's a mobile GIS tool which addresses environmental and spatial data across all regions. Access is granted to a user after correct login details are submitted.

Mobile GIS makes the data collection process faster and more efficient. The application has features such as layer selection, feature identification, search engine, feature details editing, layer switching, online map viewing and geolocation. The mobile application makes use of the device's GPS in order to tell the user's current location. In addition, the application makes use of the device's Camera to attach pictures to a location as well as the feature details.

Note:

This user guide is a working progress and more information will be added in near future.

The Mobile GIS application is mainly used for mapping and data collection. You can also use the mobile application to:

- View and navigate the map and its features
- Download/Synchronize with the main server
- Collect new GIS features
- Inspect and record the state of features and assets
- Search and organize map features requiring field visits
- Update existing GIS feature attributes by creating memos

An important advantage of using this mobile application is the ability to receive and send data to the GIS Server on ZETDC network and the ability to access data when out in the field. This ability to synchronize information enables everyone to have a common dynamic view of the latest information. Field work is accomplished in the application through the use of the Memos module.

For example, if your job is to collect features, location and status of a transformer installed in a new substation, the Collect Features task will guide you through the process of picking the transformer feature type, collecting its location using the GPS of the mobile device and setting status information etc.

Throughout the course of using the application, you are presented with helpful information and a consistent set of pages and menus to guide you through your field work.

HOW TO INSTALL

Mobile GIS is a hybrid application that looks and feels just like a native mobile app. A hybrid application is a type of mobile app that uses a browser window to display its interface. Download the apk executable file for Android named `zetdc-gis.apk` on the GIS desktop application on 172.20.25.25 and install on the mobile device. It creates an icon, like the one below, which will be used to access the application.



MobileGIS icon on mobile device

CONFIGURING WIFI SETTINGS

The steps involved in configuring WiFi settings once connected to Zetdc network

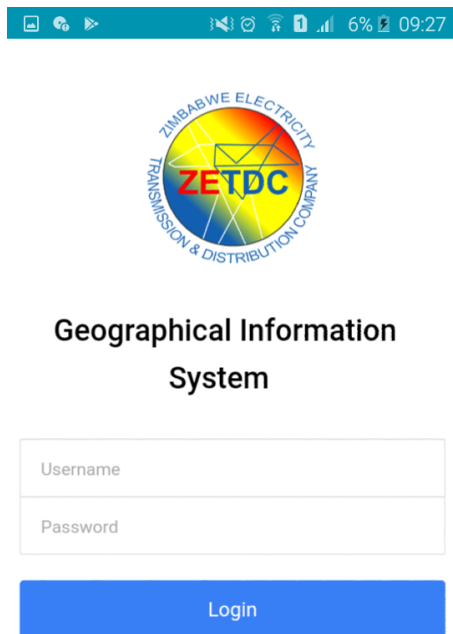
- Connect to ZETDC WiFi and enter password
- Go to settings on your mobile device
- Select WiFi
- Select the WiFi that the device is connected to and choose Modify network
- Click show advanced options
- Change proxy settings from None to Manual and enter proxy server address, port number and enter GIS server address where its written bypass proxy for
- Save settings and log in to the application

STEPS FOR SYSTEM FUNCTIONS

Mobile GIS is a user friendly and easy to use application. The steps to use the application are explained in detail as follows:

Upon opening of the application, the system checks if a session table exists inside the local database. If it exists the system goes straight to the main menu but if not it takes the user to the login page.

Logging in

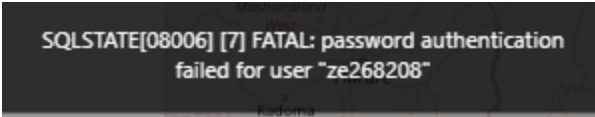


The screenshot displays the login interface of the ZETDC Mobile GIS application. At the top, there is a status bar with various icons and the time 09:27. Below this is the ZETDC logo, which is a circular emblem with a yellow and blue gradient, featuring the letters 'ZETDC' in the center. The text 'ZIMBABWE ELECTRICITY TRANSMISSION & DISTRIBUTION COMPANY' is written around the perimeter of the logo. Below the logo, the title 'Geographical Information System' is centered. Underneath the title is a login form consisting of two input fields: 'Username' and 'Password'. Below these fields is a blue button labeled 'Login'.

Login page

- User should enter the correct Username and Password in-order to have access to the system.

- Failure to provide the correct details displays an error message and denies access to the system like the one below.

A screenshot of a terminal window with a black background and white text. The text displays an SQL error message: 'SQLSTATE[08006] [7] FATAL: password authentication failed for user "ze268208"'.

```
SQLSTATE[08006] [7] FATAL: password authentication  
failed for user "ze268208"
```

Map Screen



1. Navigation menu – The navigation menu slides in from the left and allows navigation between major modules of the application. This side menu is the primary means of navigating between pages in the application, and performing the actions required to complete field tasks.

2. Main bar – displays the name of the page currently opened. In this case it is the Home Page. Since tasks contain a series of nested pages, knowing which page you are viewing at any given time is extremely important. The menu bar displays both the name of the page you are currently viewing and a graphical icon representing the task that it is contained within.

3. Geolocator – identifies the actual geographical location of the person by using the GPS of the mobile device.

4. Zoom control – used to change the zoom levels on the map. + zooms in for a closer look and – zooms out of the map.

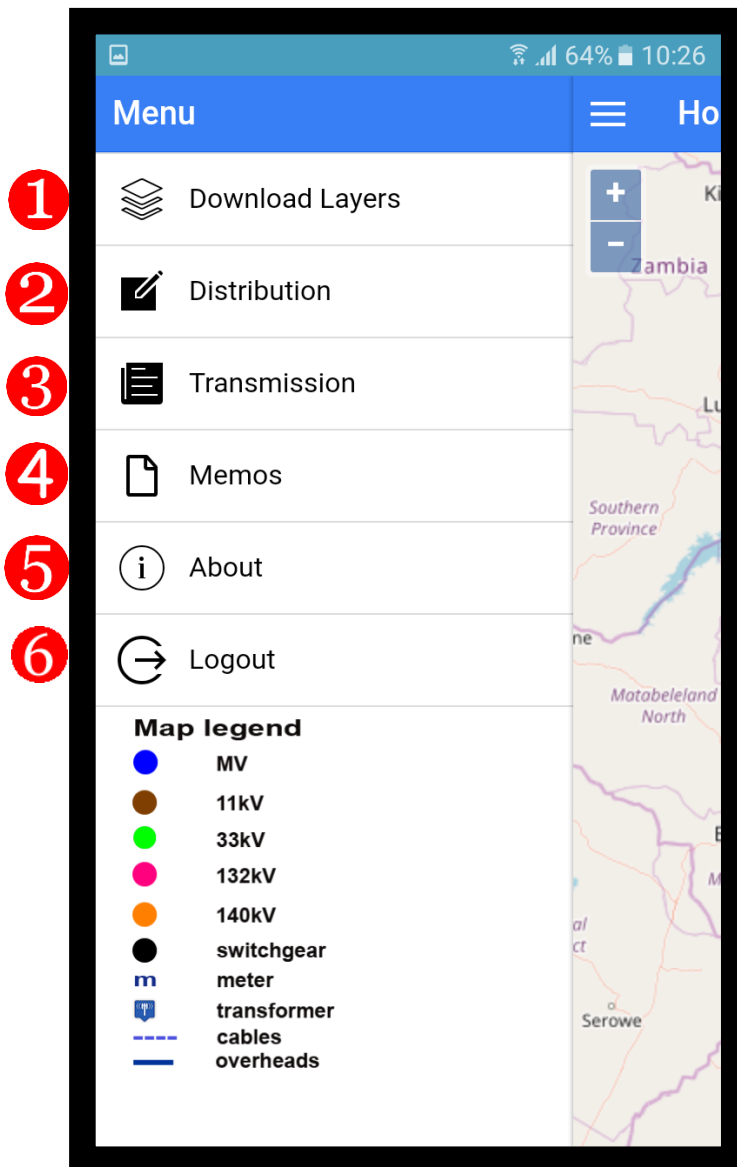
5. Base Map – base layer with the map.

When you take Mobile GIS to the field, your downloaded features are stored on your mobile device—either on a storage card or in main memory.

6. Search

User can search for a particular feature on the map using feature name.

Side menu



1. **Download Layers** - This button allows a user to go to the layer selection interface whereby a user can select which features they want to work on and download them from the main GIS server. *described on page 7.*

2. **Distribution** - Used in the field to collect information about a new feature under the Distribution Layer. Clicking this button allows a user to start creating a new memo on a new feature which is not already in the system.

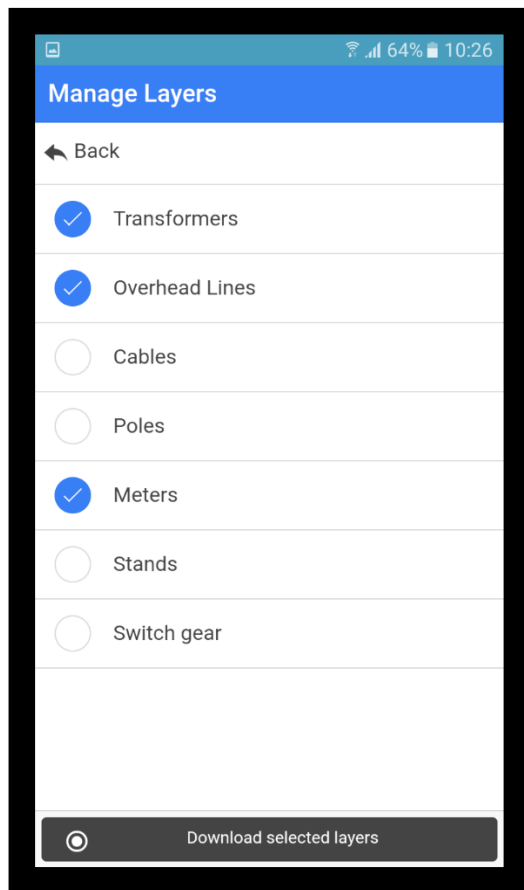
3. **Transmission** – This Layer allows the user to capture features in the Transmission Layer

4. **Memos** - Allows a user to view the list of memos created whilst in the field and currently stored on the local storage of the mobile device.

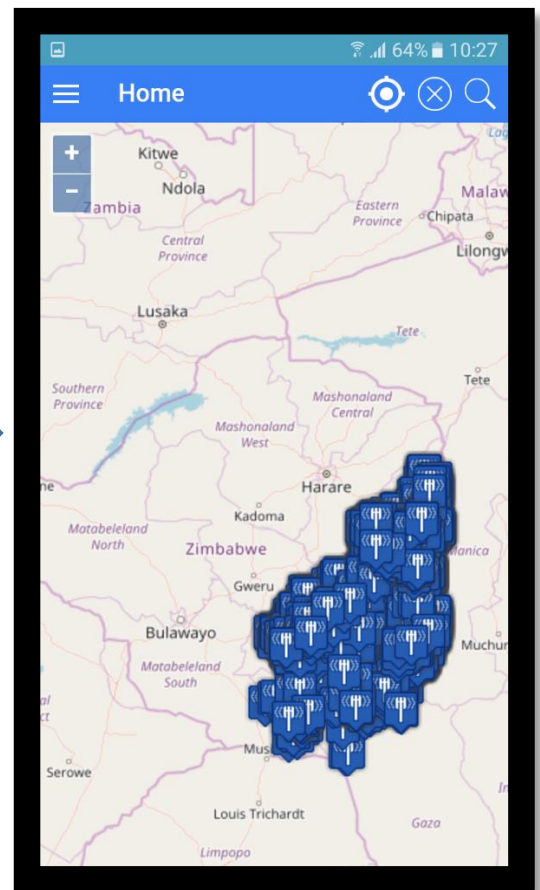
5. **About** - Shows the information about the application.

6. **Logout** - Clicking this button will log the user out of the application. In-order to re-gain access to the application, the user has to log in with correct credentials again.

Layer selection

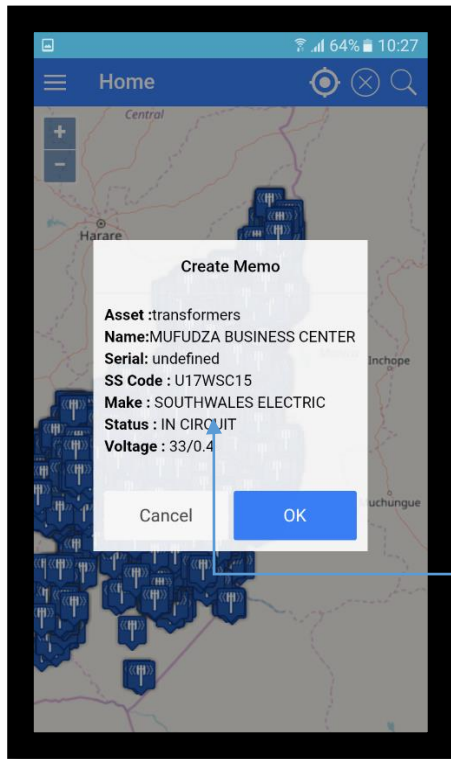


Selected layers to be downloaded from Server



Map populated with features downloaded from Server

- Layer Selection window appears after selecting the Sync layers button on the side menu.
- This interface allows a user to select the features that they wish to work on.
- By clicking the Sync Selected Layers button at the bottom, the selected layers are downloaded from the main server into the mobile device.
- The back button allows the user to go back to the main menu where the features will be populated on the map with their exact positions on the ground.



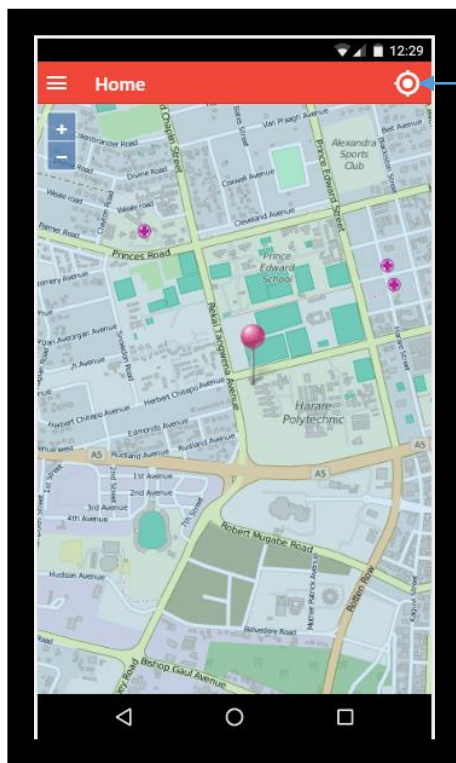
Memo creation

Map showing Transformers, Overhead lines and Meters downloaded into the mobile device. Features can be identified by clicking on the feature and a pop up appears with a brief information about the feature.

To get more information click OK otherwise Cancel.

Pop up to create Memo on a transformer

Geolocation

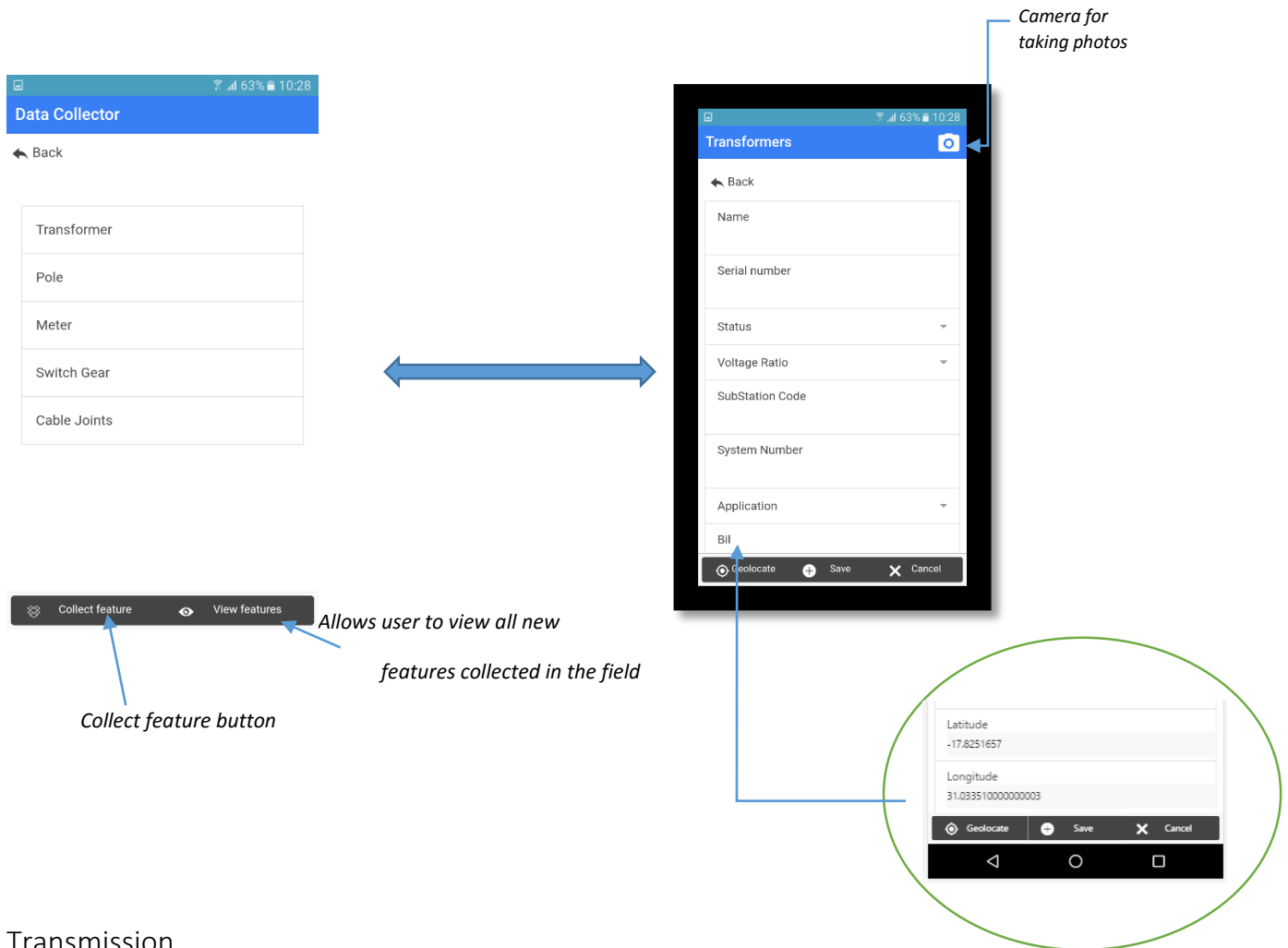


Geo-location icon

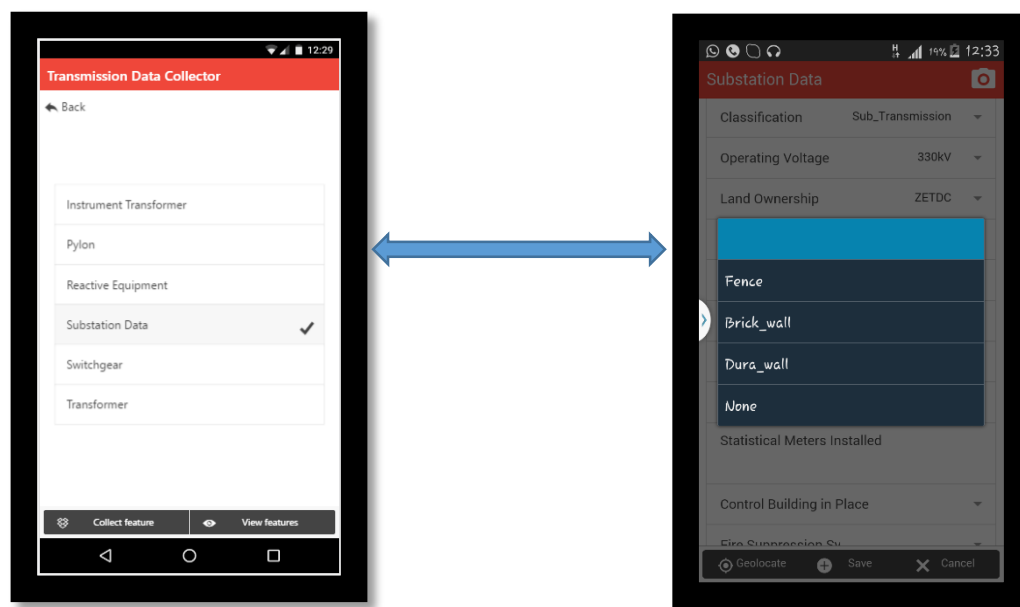
My current location

Clicking the Geo-locator icon at the top right of the page locates the user's current position using the GPS of the mobile phone and displays it with the icon as shown in the image.

Distribution



Transmission



See Data Collection module page 12

Data Collection

Data collection module allows a user to collect information of a new feature whilst in the field.

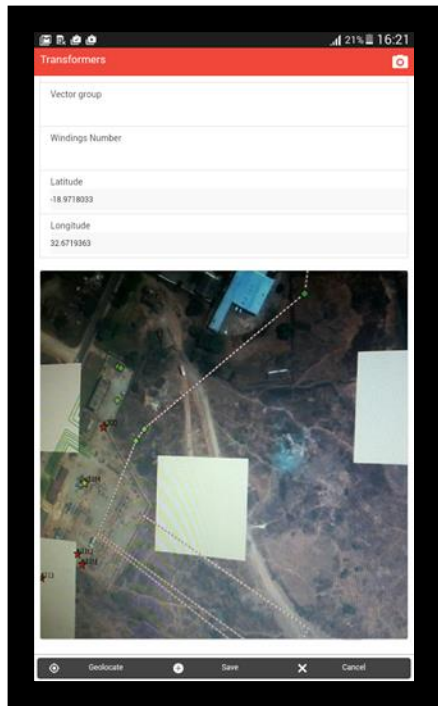
Diagram 1 shows the interface where one selects the feature that they want to collect in the field. Clicking Collect feature button opens a form with the attributes of the feature selected as shown in diagram 2.

The form allows a data collector to input data in the fields and also to take a picture of the feature by clicking the camera icon at the top left corner of the form. The image will be attached inside the form.

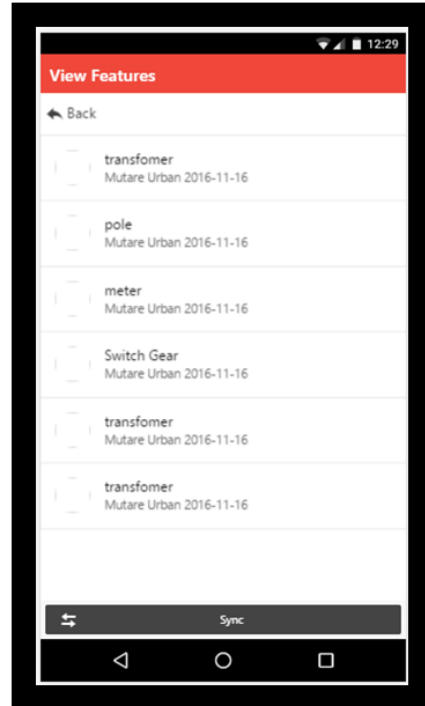
Other information which can be collected is the actual coordinated of the current position which is done by clicking the Geo-locate button at the bottom. After clicking the Geo-locate button, it automatically populates the latitude and longitude on the form. As shown above.

The cancel button clears all the information entered inside the fields within the form. The back button at the top right corner is used to go back to the previous page if one once to select a different feature to collect.

The save button allows the user to save the work done. A pop up message appears to confirm saving data in the local storage of the mobile device. In-order to view these forms and the information created on these features, the user has to select Memos in the side menu of the application.

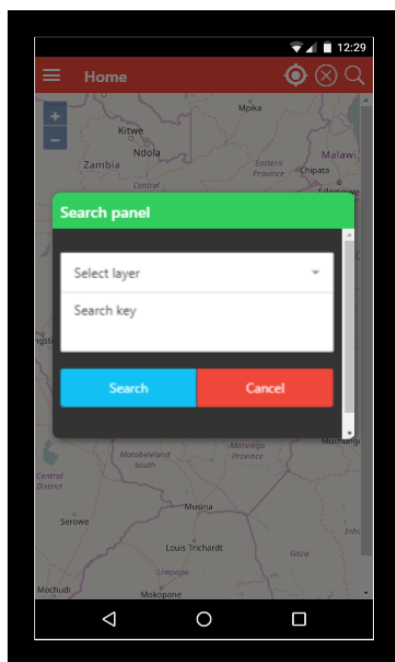


Camera allows user to take photos and preview the photo of the feature being collected.



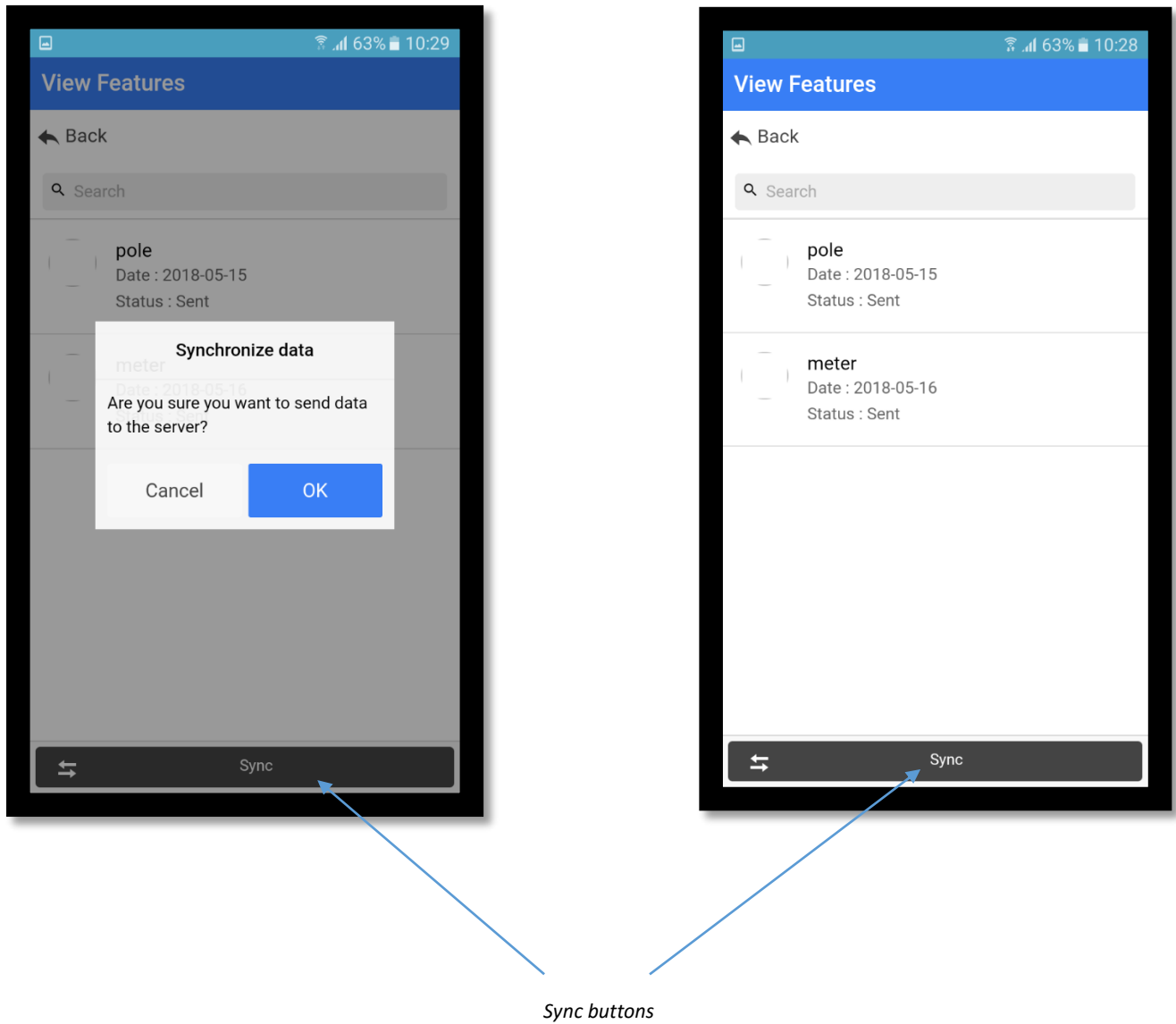
Memos and features created in the field can be in a list before synchronizing with main server

Search



Search allows user to search a feature on the map using the feature name. On clicking search button, it zooms in to the location of the feature being searched and points out the feature.

Synchronizing with main server



Sync allows user to synchronize/send locally stored data to the main GIS server.

About



The About page in this application shows information about the system such as:

- i. Name of the application
- ii. Version of the application
- iii. Information about the application
- iv. Help link, which opens the browser for more information about the application.

Acronyms and Abbreviations

GIS:	Geographical Information System
Sync:	Synchronize with the main server
GPS:	Global Positioning System